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Lattice Boltzmann Method for high Reynolds number compressible flow

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A B S T R A C T

We employ a Lattice Boltzmann Method (LBM) for modelling high Reynolds number (*Re*) compressible supersonic flow using D2Q9 lattice. In this model, a two-population thermal lattice Boltzmann (LB) is utilized in order to allow varying Prandtl number and adiabatic exponent. The equilibrium population f_i^{eq} term for mass and momentum conservation is matched analytically with order of accuracy $O(Ma^4)$, while the equilibrium population g_i^{eq} term for energy conservation is solved numerically through Lagrange multipliers method. By using more accurate equilibrium population g_i^{eq}, the present numerical method shows the enhancement of stability, especially at high Reynolds and Mach number flows. To simulate high Reynolds and Mach number (*Ma*) flows efficiently, we adopt a shifted lattice scheme, stabilized by relaxation time depending on the Knudsen number. The LBM show excellent accuracy through 1D Riemann examples, capable of handling expansion shock waves and discontinuity. Our results for supersonic flow past a cylinder also find excellent agreement with experimental data in literature. Separately, we benchmark results for transonic and supersonic flows past a NACA0012 aerofoil as a demonstration of its simplicity, versatility for applications involving high Reynolds and Mach numbers ($Ma_\infty = 1.5,\ Re = 10^7$).

1. Introduction

Recently, the Lattice Boltzmann Method (LBM) has received significant attention due to widespread applications in complex fluid flows such as rarefied gas [1] and multiphase [2] flows involving heat transfer [3], turbulence [4] and porous media [5]. The LBM has demonstrable advantages in simplicity of implementation and facilitating to use on GPU [6]. The LBM originates from the kinetic theory of gases in the form of the Boltzmann equation implemented on a discrete lattice domain, whose collision term is simplified by the Bhatnagar, Gross and Krook (BGK) model [7] and the velocity distribution function representing discrete populations of molecules at discrete velocities. The Navier-Stokes equations is reproducible in hydrodynamic limit via Chapman-Enskog expansion [8].

Although the LBM enjoys considerable success in simulating incompressible fluid flow [9], there are certain limitations under high Reynolds and Mach number flows. First, the common used population term $\left(f_i^{eq} = \omega_i \rho\left(1 + \frac{\vec{e_i}\vec{u}}{c_s^2} + \frac{(\vec{e_i}\vec{u})^2}{2c_s^4} - \frac{\vec{u}^2}{2c_s^2}\right) + O(Ma^3)\right)$ is the second order accuracy in Mach number. Second, the relaxation time $(\tau = 0.5 + \nu/c_s^2)$ will be close to 0.5 under high Reynolds numbers, leading to instability and divergence of numerical solutions. Third, since the conventional LBM is based on static velocity discretization, out of equilibrium effects including sudden large perturbations of macroscopic quantities can introduce significant deviations from actual physics [10]. This limitation can be mitigated using larger stencils, which can increase computational costs.

For high Mach number flows, Frapolli et al. [11] proposed the space-filling shifted lattices method with co-moving Galilean reference frame with shifted discrete velocity set. Hosseini et al. [12] also performed an extensive analysis of the Lattice Boltzmann Method on shifted stencils. This opens a new space to explore the subsonic, transonic and supersonic flows, with added costs in terms of computational overhead in parallel simulations [11,13]. In order to overcome the limitation of high Reynolds number flow, Karlin et al. [14] proposed the entropic LBM, which uses the positive root of the entropy condition as a factor of relaxation parameter. However, the method is only applicable when the root of entropic condition exists in the first place. For numerical stabilization, Coreixas et al. [10] proposed a Knudsen-number-dependent relaxation time method where the relaxation time correction coefficient is an increasing piecewise constant function of the local Knudsen number. A review of stabilization methods including Multiple Relaxation Time (MRT), entropic LBM and filtering method can be found in Ref. [15].

Here, we review recent progress in LBM simulations of supersonic flow. Farag et al. [16] introduced new pressure-based Lattice-Boltzmann method (HRR-p) for the simulation of flows for Mach numbers ranging from 0 to 1.5. The model is tested in a one-dimensional shock tube and a two-dimensional shock–vortex interaction. Guo et al. [17] developed an improved thermal hybrid recursive regularized lattice Boltzmann

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(iHRR-ρ) model for compressible flows, validated using a test case of supersonic flow over a NACA aerofoil at $Re = 10^4$ and $Ma = 1.5$. Feng et al. [18] proposed a hybrid recursive regularized D3Q19 LBM (HRR-LBM) model with applications to compressible aerodynamics and focusing on efficient boundary conditions. Zhong et al. [19] proposed a discrete unified gas-kinetic scheme for modelling compressible flows at high Mach numbers with results in agreement with test cases. Saadat et al. studied compressible flows based on standard lattice [20] and unstructured mesh [21], as well as shifted lattice with additional correction terms to satisfy the third and fourth Maxwell-Boltzmann moments in order to recover fully Navier-Stokes-Fourier equations [22–24]. Since the equilibrium population g_i^{eq} term is only second order accurate, however, numerical stability under for very high Reynolds and Mach numbers is not guaranteed. Saadat et al. [25] proposed the full expansion of the equilibrium population g_i^{eq} term for compressible flow simulations. Latt et al. [26] reported the LBM simulations for supersonic flow by using equilibrium population f_i^{eq} optimized through Lagrange multipliers [27–30] and Knudsen-number-dependent relaxation time stabilizing methods. The simulation may carry high computational costs due to Lagrange multipliers method with 13 constraints. Hosseini et al. [31] performed LBM simulations for compressible supersonic flows with $Ma = 1.5$ over a NACA0012 aerofoil at Reynolds numbers of only 2000. Frapolli et al. [30] used high order lattices entropic LBM. The equilibrium population f_i^{eq} is numerically calculated using Lagrange multipliers method with 5 constraints. However, the usage of high order lattices can lead to the high computational cost, especially for three-dimensional simulations. For all work above, the highest reported Reynolds number, for example of supersonic flow ($Ma = 1.5$) passing NACA0012 aerofoil, is only $Re = 10^4$. It could not meet the requirement for real supersonic flow application.

In this present work, we propose an efficient and stable LBM for high Reynolds and high Mach number flows which can simulate the supersonic flow with Reynolds number to $Re = 10^7$. We use Lagrange multipliers method to obtain the accurate equilibrium population g_i^{eq} and analytical fourth order accuracy f_i^{eq} term in order to avoid the approximation error. We also adopt a shifted lattice scheme, Knudsen-number-dependent relaxation time stabilizing technique to enhance the numerical stability. Our LBM methodology also takes advantage of computation cost due to using the conventional D2Q9 lattice instead of a higher order lattice [30] and using 3 equations of constraint in Lagrange multipliers method instead of 13 equations [26]. We also perform the benchmarking with various examples for expansion shock wave in 1D Riemann, supersonic flow over cylinder, transonic and supersonic flows over the NACA0012 aerofoil to consolidate the proposed method.

For organization, the numerical method is described in Section 2. Some numerical examples are discussed in Section 3 and we summarize our work in Section 4.

2. Numerical method

We use a two-population LBM to allow varying Prandtl number and adiabatic exponent. The kinetic equations [28] are

$$f_i(x + c_i\delta t, t + \delta t) - f_i(x,t) = \omega(f_i^{eq} - f_i), \tag{1}$$

$$g_i(x + c_i\delta t, t + \delta t) - g_i(x,t) = \omega(g_i^{eq} - g_i) + (\omega - \omega_T)\left(g_i^{*} - g_i^{eq}\right), \tag{2}$$

where ω and ω_T are relaxation factors related to the dynamic viscosity and thermal conductivity. f_i and f_i^{eq} are, respectively, the population distribution function and the local equilibrium distribution function for mass and momentum conservation. g_i, g_i^{*} and g_i^{eq} are, respectively, the population, quasi and local equilibrium distribution functions for energy conservation.

The dynamic viscosity and thermal conductivity are, respectively,

$$\mu = \left(\frac{1}{\omega} - \frac{1}{2}\right)\rho T, \tag{3}$$

and

$$\kappa = C_p\left(\frac{1}{\omega_T} - \frac{1}{2}\right)\rho T. \tag{4}$$

Both the Prandtl number $Pr = C_p\mu/\kappa$ and the adiabatic exponent $\gamma = C_p/C_v$ are adjustable, where C_v is the specific heat at constant volume and $C_p = C_v + 1$ is the specific heat at constant pressure.

For high Mach number flows, we adopt the concept of shifted lattices [11] where under the moving reference frame, the discrete velocity vector is

$$c_i = v_i + U, \tag{5}$$

where $U = (U_x, U_y)$ is the moving reference frame velocity and $v_i = (v_{ix}, v_{iy})$ is the standard discrete velocities. For D2Q9 lattice,

$$v_i \in \begin{bmatrix} 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 & 0 \end{bmatrix}. \tag{6}$$

To satisfy mass, momentum and energy conservation laws, we define

$$\rho = \sum_{i=1}^{Q} f_i = \sum_{i=1}^{Q} f_i^{eq}, \tag{7}$$

$$\rho u_\alpha = \sum_{i=1}^{Q} c_{i\alpha} f_i = \sum_{i=1}^{Q} c_{i\alpha} f_i^{eq}, \tag{8}$$

$$2\rho E = \sum_{i=1}^{Q} g_i = \sum_{i=1}^{Q} g_i^{eq}, \tag{9}$$

where ρ is the fluid density, ρE is the total fluid energy, $u_\alpha (\alpha = x, y)$ is the fluid velocity, Q is the number of discrete velocity vectors and $c_{i\alpha}$ is the lattice velocity.

The fluid temperature is

$$T = (1 / C_v)\left(E - \boldsymbol{u}^2 / 2\right) \tag{10}$$

where $\boldsymbol{u}$ is the fluid velocity magnitude.

To recover exactly the Navier-Stokes-Fourier equations in hydrodynamic limit, the local equilibria must not only satisfy the conservation laws but also the second, third and fourth orders of Maxwell-Boltzmann moments [20]. However, resolving higher moments comes with commensurate computational costs and numerical instability issues. The Chapman-Enskog expansion yields a minimal set of 8 constraints in two dimensions [10] (13 in three dimensions [26]) for the simulation of convective phenomena. This 8-constraint model satisfies mass, momentum, energy conservation laws, second and trace of third-order Maxwell-Boltzmann moments. In our model, we use Eq. (9) for equilibria g_i^{eq}, hence, the 8-constraint model [10] is equivalent to the model satisfying conservation laws Eqs. (7)–((9)) and Maxwell-Boltzmann relations below [20]

$$P_{\alpha\beta}^{eq} = \sum_{i=1}^{Q} c_{i\alpha} c_{i\beta} f_i^{eq} = \rho u_\alpha u_\beta + \rho T \delta_{\alpha\beta}, \tag{11}$$

$$q_\alpha^{eq} = \sum_{i=1}^{Q} c_{i\alpha} g_i^{eq} = 2\rho u_\alpha (E + T), \tag{12}$$

where $\delta_{\alpha\beta}$ is the Kronecker delta.

The difference between quasi and local equilibrium distribution functions is [20]

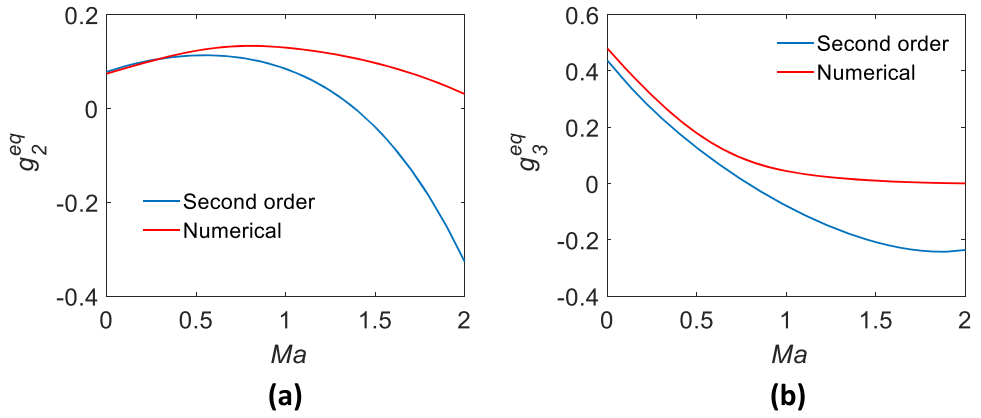


Fig. 1. Equilibrium populations estimated by second order approximation [20] and present numerical calculation as a function of Mach number. (a) g_2^{eq} and (b) g_3^{eq}.

$$g_i^* - g_i^{eq} = W_i \frac{2u_\beta \left(P_{\alpha\beta} - P_{\alpha\beta}^{eq}\right) c_{i\alpha}}{T}, \tag{13}$$

where

$$P_{\alpha\beta} = \sum_{i=1}^{Q} c_{i\alpha} c_{i\beta} f_i. \tag{13a}$$

The temperature-dependent weights W_i are

$$W_i = W_{ix} W_{iy}, \tag{14}$$

where

$$W_{-1} = T/2, \tag{14a}$$

$$W_0 = 1 - T, \tag{14b}$$

$$W_{+1} = T/2. \tag{14c}$$

For closure, the equilibrium distribution functions g_i^{eq} and f_i^{eq} must be accurately evaluated. The equilibrium distribution function f_i^{eq} is rendered fourth order accuracy $O(Ma^4)$ through minimization of the entropy function $S_f = \sum_{i=1}^{Q} f_i \ln(f_i / W_i)$, subjected to the local conservation laws Eqs. (7), (8) and the equilibrium pressure tensor (Eq. (11)) [22] as

$$f_i^{eq} = \rho \Phi_{ix} \Phi_{iy}, \tag{15}$$

where

$$\Phi_{-1} = \frac{-(u_\alpha - U_\alpha) + (u_\alpha - U_\alpha)^2 + T}{2}, \tag{15a}$$

$$\Phi_{+1} = \frac{(u_\alpha - U_\alpha) + (u_\alpha - U_\alpha)^2 + T}{2}, \tag{15b}$$

$$\Phi_0 = 1 - \left[(u_\alpha - U_\alpha)^2 + T\right], \tag{15c}$$

where $\alpha = x, y$.

As for equilibrium distribution function g_i^{eq}, instead of using second order accurate approximations [20], it is numerically calculated. More accurate solutions of g_i^{eq} contribute towards stable and accurate simulations, especially for high Reynolds and Mach number flows. As with f_i^{eq}, g_i^{eq} is obtained by minimizing the entropy function $S_g = \sum_{i=1}^{Q} g_i \ln(g_i / W_i)$, constrained by energy conservation (Eq. (9)) and trace of third-order Maxwell-Boltzmann moment (Eq. (12)). Lagrange multipliers method is then applied, and the solution has the form as

$$g_i^{eq} = \rho W_i \exp(\chi + \xi_\alpha c_{i\alpha}) \tag{16}$$

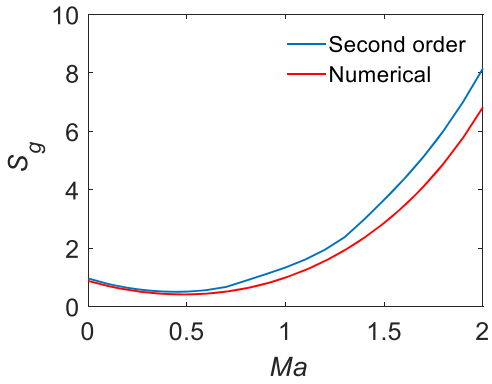


Fig. 2. Entropy function $S_g = \sum_{i=1}^{Q} g_i \ln(g_i / W_i)$ estimated by second order approximation[20] and present numerical calculation as a function of Mach number.

By substitute Eq. (16) into Eqs. (9) and (12), we have a set of equations where the solution of Lagrange multipliers χ and ξ_α are numerically obtained by using Newton-Raphson method. Here, our LBM is physically equivalent to 8-constraint model [10]. However, the equilibrium distribution function f_i^{eq} is already analytically approximated by Eq. (15). Hence, we only need to solve 3 equations of constraint for g_i^{eq} instead of 8 as Coreixas's work [10] for two-dimensional simulation.

In order to point out a remarkable property of the numerical equilibrium, g_2^{eq} and g_3^{eq} estimated by second order approximation [20] and present numerical calculation are plotted in Fig. 1 as a function of Mach number with conditions of shifted lattice velocity $U = (0.5,\ 0)$ and $T_\infty = 0.25$. As observation, both second order approximation equilibrium populations are close to numerical values when Mach number is less than 0.5. However, they are diverged from numerical one when Mach number increases. Furthermore, they become negative after a certain value of Mach number, violating the positivity constraints imposed by the construction of the equilibrium [30], meanwhile the numerical solutions are still positive. Fig. 2 shows the entropy function S_g with respect to Mach number. It is obvious that the entropy function estimated by second order is not the minimum, especially when Mach number is larger than 0.5. It is possible to conclude that the numerical estimation of g_i^{eq} is not only more accurate but also enhances the stability of the simulation.

One observes in Fig. 1b a discrepancy between the second order and numerical solutions for g_3^{eq} under low Mach numbers. The second order formula satisfies energy conservation (Eq. (9)) and Maxwell-Boltzmann relation (Eq. (12)), but not the minimum entropy function, as shown in Fig. 2, where the observed difference between the second order and

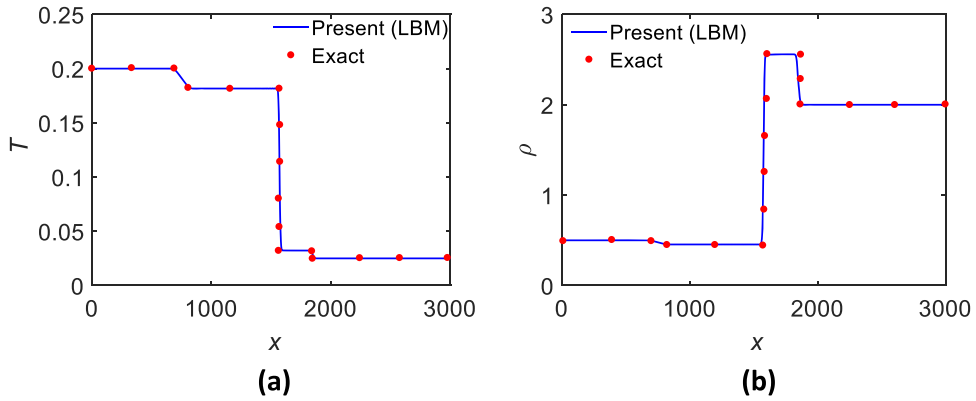


Fig. 3. Comparison of gas properties along the Sod shock tube between present LBM result and exact solution [20]. (a) Temperature and (b) Density.

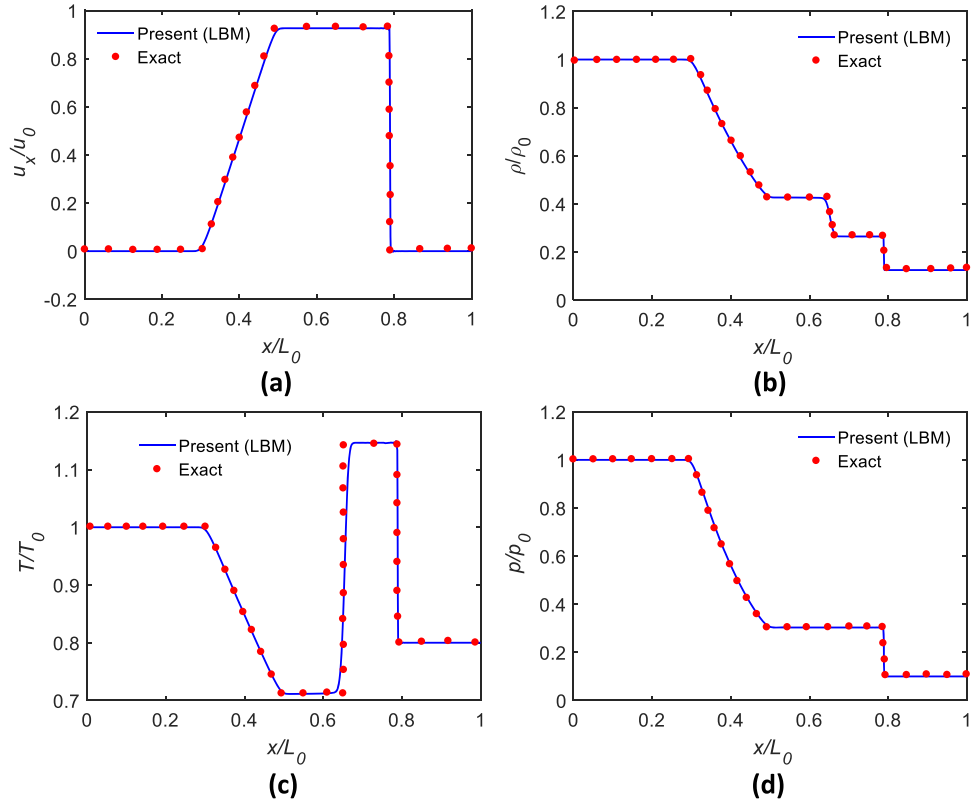


Fig. 4. Comparison of non-dimensional gas properties along the Sod shock tube between present LBM result and exact solution [32]. (a) Velocity, (b) density, (c) temperature and (d) pressure.

numerical solutions for S_g correlates with that for g_3^{eq}.

3. Results

In this section, we benchmark and validate our LBM model based on case studies, including shock and expansion wave, transonic, supersonic, low and high Reynolds number flows. Unless otherwise stated, the default Prandtl number is $Pr = 0.71$ and the adiabatic exponent is $\gamma = 1.4$. The current LBM method is capable of simulating supersonic flows up to a far-field Mach number $Ma_\infty = 2$.

3.1. Sod shock tube

The Sod shock tube is a classical test case for Riemann solvers. It is a good example to test the compressible flow code because it includes a shock wave, a contactless surface, and an expansion wave. Here, we consider two different cases.

Case 1. Sod shock tube with initial condition,

$$(\rho, u_x, u_y, T) = \begin{cases} (0.5,\ 0,\ 0,\ 0.2), & x/L_x \le 0.5, \\ (2.0,\ 0,\ 0,\ 0.025), & x/L_x > 0.5. \end{cases} \tag{17}$$

A grid of 3001×5 for computational domain, the viscosity $\mu = 0.025$, specific heat ratio $\gamma = 2.0$ and zero shifted lattice velocity are used for the simulation. Fig. 3 shows the temperature and gas density along the Sod shock tube at time $t = 1273$ in the lattice units. Our LBM results are in excellent agreement with exact analytical solutions [20].

Case 2. Sod shock tube with initial condition as

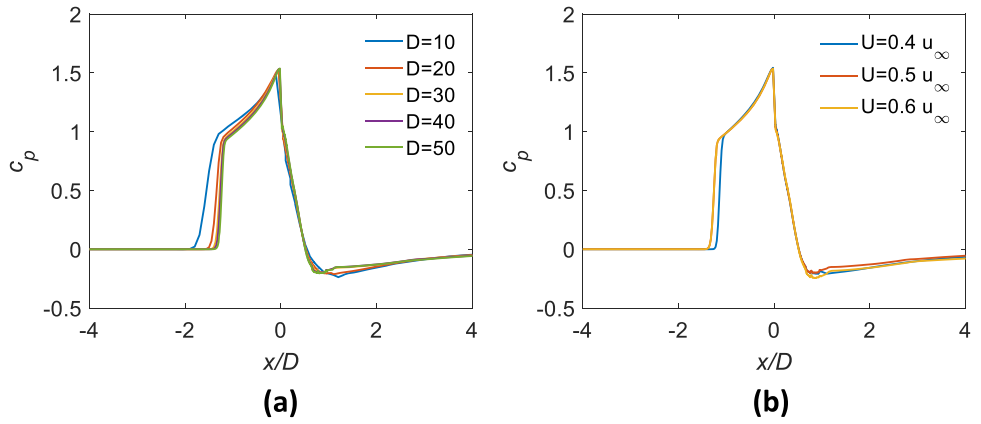


Fig. 5. Simulated pressure coefficients around the cylinder surface for a supersonic flow $Ma_\infty = 1.6,\ Re = 300$. (a) Various number of grid points along the diameter of cylinder and (b) various shifted velocity. x/D=0 position is at the cylindrical head.

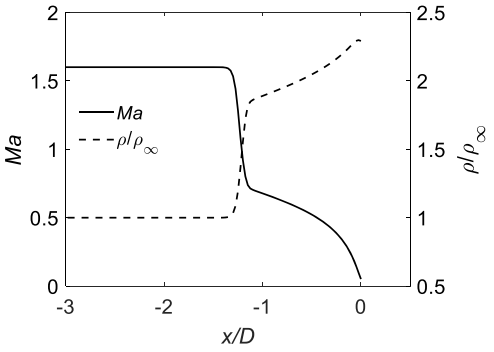


Fig. 6. Mach number and density ratio of the flow along the stagnation streamline for a supersonic flow across a circular cylinder $Ma_\infty = 1.6,\ Re = 300$.

$$(\rho/\rho_0, u_x/u_0, p/p_0) = \begin{cases} (1,\ 0,\ 1), & 0 < x/L_0 \le 0.5, \\ (0.125,\ 0,\ 0.1), & 0.5 < x/L_0 < 1. \end{cases} \tag{18}$$

Here, $\rho_0 = 1$, $p_0 = 0.2$, $T_0 = 0.2$, $u_0 = \sqrt{0.2}$ and $L_0 = 1501$ are in lattice units. A grid of 1501×5 for computational domain, a low viscosity $\mu = 10^{-4}$, specific heat ratio $\gamma = 1.4$ and the shifted lattice velocity $U = (0.4,\ 0)$ are used for the simulation. Fig. 4 shows the non-dimensional velocity, density, temperature and gas pressure along the Sod shock tube at time $t = 552$. Again our LBM results are in excellent agreement with exact analytical solutions [32], with only minor deviations in Fig. 4**c**.

3.2. Supersonic flow across a circular cylinder

Supersonic flow across a circular cylinder is numerically simulated by using current LBM method. Pressure coefficients around the circular cylinder are plotted in Fig. 5**(a)** for various grid resolutions expressed in terms of number of grid points along the diameter of cylinder, $D = \{10,\ 20,\ 30,\ 40,\ 50\}$. Simulation parameters are the Reynolds number $Re = 300$ with cylinder diameter as characteristic length, the far-field Mach number $Ma_\infty = 1.6$, the far-field temperature $T_\infty = 0.25$ and the shifted lattice velocity $U = (0.5u_\infty,\ 0)$. The cylinder is centered at a horizontal position $5D$ within a square computational domain of size $15D \times 15D$ with uniform Cartesian mesh. Dirichlet boundary condition applies on the inlet left boundary for velocity, temperature and air density, Neumann condition on the outlet right boundary for velocity, temperature, density and symmetry conditions on upper and lower boundaries. On the surface of cylinder, the boundary conditions are no-slip for flow and no-penetration for temperature and density [21]. A separate mesh independence study shows that a grid solution of $D = 40$ grid points is sufficient for this problem.

Fig. 5**(b)** shows the pressure coefficients around the cylinder surface for various shifted lattice velocities $U = \{0.4u_\infty,\ 0.5u_\infty,\ 0.6u_\infty\}$. The overall results show weak dependence on the shifted lattice velocity, except for the shock front and wake behind the cylinder. In the present case, we found numerical convergence only within the range of $0.4u_\infty \le U \le 0.6u_\infty$, where $U = 0.5u_\infty$ is set as the base lattice velocity for subsequent study.

Fig. 6 shows the Mach number and density ratio profiles along the stagnation streamline for a far-field Mach number $Ma_\infty = 1.6$. We observe a sudden fluctuation in Mach number and flow density at the

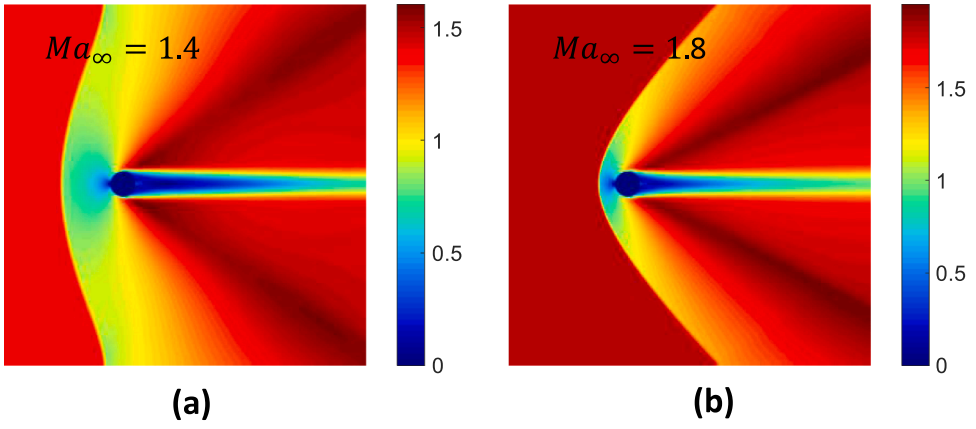


Fig. 7. Mach number field for supersonic flow around a cylinder for $Re = 300$ for far-field Mach numbers (a) $Ma_\infty = 1.4$ and (b) $Ma_\infty = 1.8$.

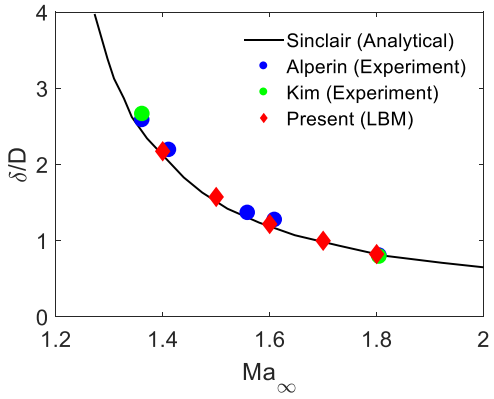


Fig. 8. Comparison of modelled shock stand-off distance δ/D with respect to far field Mach number Ma_∞ against reported experimental [33,34] and analytical solutions [33].

shock front, and stopping flow at the head of cylinder (x/D=0). Fig. 7 shows steady-state Mach number fields under two far-field Mach numbers, $Ma_\infty = 1.4$ and $Ma_\infty = 1.8$. The simulation visually captures the bow shock ahead of the cylinders and how stand-off distance decreases with increasing far-field Mach numbers, a trend corroborated by reported experimental and analytical data. Fig. 8 shows the comparison between the current simulation, analytical [33] and experimental results [33,34] of the shock stand-off distance δ/D, which is the distance of the shock front from cylinder δ normalized by the cylinder diameter D. The good agreement further affirms the validity of our LBM methodology.

3.3. Transonic flow past NACA0012 aerofoil

We examine transonic flow past a NACA0012 aerofoil for far-field $Ma_\infty = 0.8$ and $Ma_\infty = 0.9$ at $Re = 500$. Other parameters include the number of grid point along the aerofoil chord $C = 100$, far field temperature $T_\infty = 0.25$, angle of attack $\alpha = 0^o$ and shifted lattice velocity $U = (0.2u_\infty, 0)$. Uniform Cartesian mesh is used for a computational domain size $20C \times 20C$ and the aerofoil head is set at $6.75C$. The boundary conditions are same as flow over cylinder case. In the present case, numerical convergence was found for shifted lattice velocity values $0.1u_\infty \leq U \leq 1.4u_\infty$. Fig. 9 shows the Mach number distribution for both cases near the aerofoil and that the simulation result captures the features of transonic flow over the aerofoil. Fig. 10 shows a good comparison agreement between current LBM method and the numerical work of Hafez and Wahba [35] for pressure coefficient $c_p = (p - p_\infty)/(0.5\rho_\infty u_\infty^2)$.

Fig. 11 shows the Mach number and density ratio profiles of the flow along the stagnation streamline for the flow past NACA0012 aerofoil at $Re = 500$ for far field Mach numbers $Ma_\infty = 0.8$ and $Ma_\infty = 0.9$. It is observed in both cases that the Mach number decreases to zero at the aerofoil tip and the density increases along the stagnation streamline.

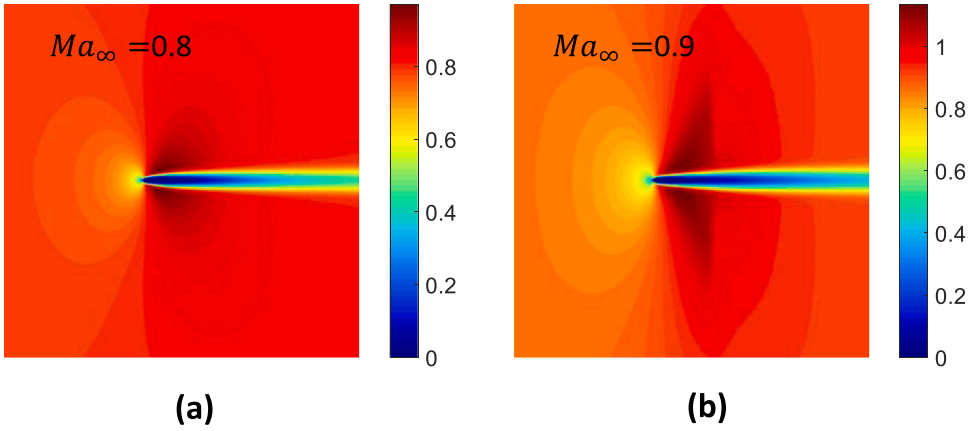


Fig. 9. Mach number distribution for transonic flow around a NACA0012 aerofoil at $Re = 500$ for far field Mach numbers (a) $Ma_\infty = 0.8$ and (b) $Ma_\infty = 0.9$.

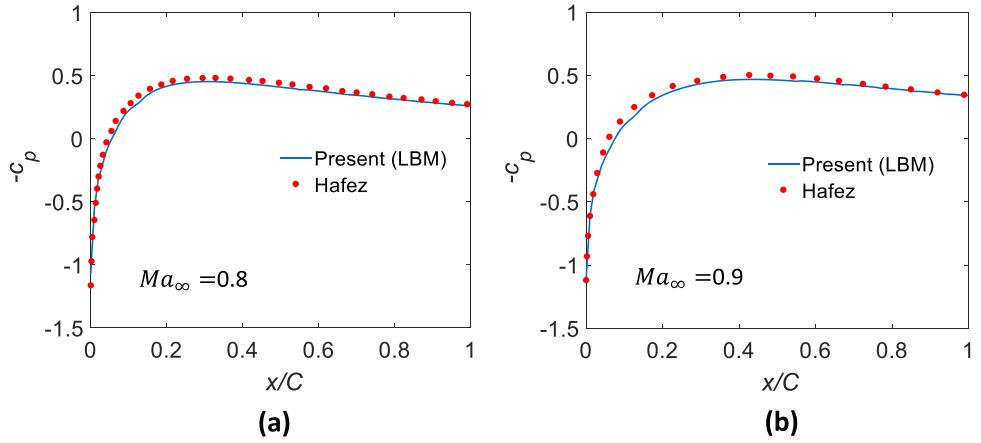


Fig. 10. Comparison of simulated pressure coefficient around NACA0012 aerofoil surface at $Re = 500$ for far field Mach numbers (a) $Ma_\infty = 0.8$ and (b) $Ma_\infty = 0.9$, with Ref. [35].

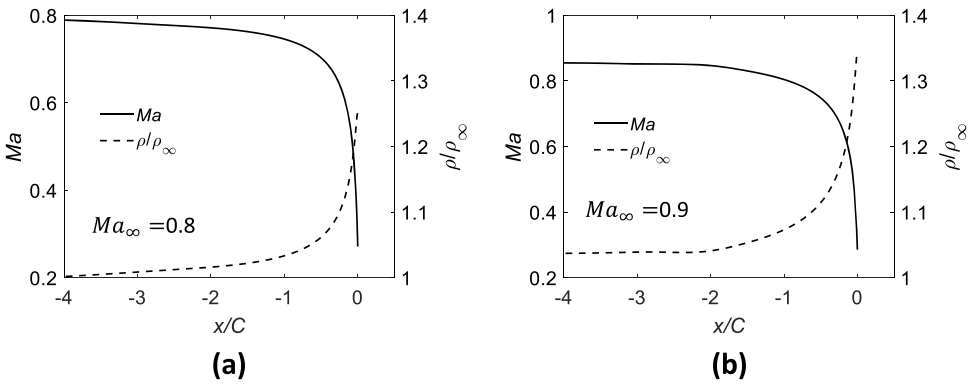


Fig. 11. Mach number and density ratio of the flow along the stagnation streamline for the flow past NACA0012 aerofoil at $Re = 500$ for far field Mach numbers (a) $Ma_\infty = 0.8$ and (b) $Ma_\infty = 0.9$.

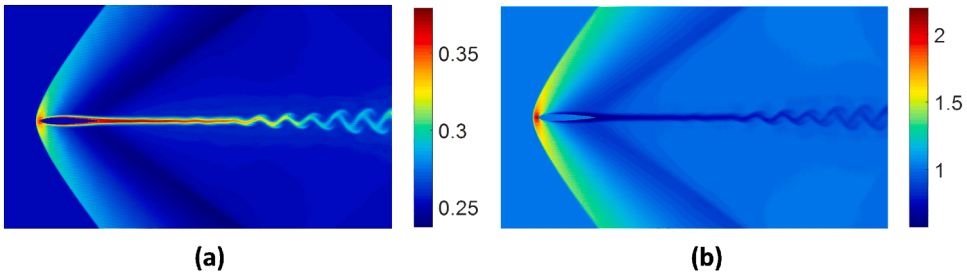


Fig. 12. (a) Temperature and (b) density fields for supersonic flow past a NACA0012 aerofoil with far-field flow of $Ma_\infty = 1.5$, $Re = 10^4$.

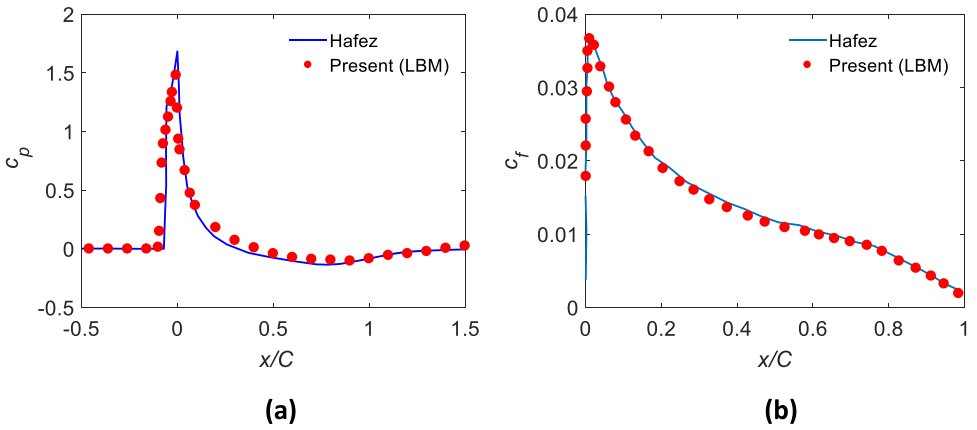


Fig. 13. Comparison of (a) simulated pressure coefficient and (b) skin friction coefficient around NACA0012 aerofoil surface ($Ma_\infty = 1.5$, $Re = 10^4$) using grid resolution $C = 150$, with Ref. [36].

3.4. Supersonic flow past NACA0012 aerofoil

Next we examine supersonic flow past a NACA0012 aerofoil for far-field $Ma_\infty = 1.5$ at $Re = 10^4$. For high Reynolds number flow, the viscosity will be small and this results in the numerical instability in the simulation. Here, we adopt the stabilization technique proposed by Latt et al. [26] and Coreixas and Latt [10], which employs a local relaxation time $\tau(\epsilon) = \tau\sigma(\epsilon)$ where τ is the global relaxation time and $\sigma(\epsilon)$ is a correction coefficient, a piecewise constant function of the local Knudsen number ϵ to dampen local perturbations. The local Knudsen number and correction coefficient are

$$\epsilon = \frac{1}{Q}\sum_{i=1}^{Q}\frac{|f_i - f_i^{eq}|}{f_i^{eq}}, \tag{19}$$

$$\sigma = \begin{cases} 1.0 & \epsilon < 10^{-2} \\ 1.05 & 10^{-2} \le \epsilon < 10^{-1} \\ 1.35 & 10^{-1} \le \epsilon < 1.0 \\ 1/\tau & 1.0 \le \epsilon \end{cases}. \tag{20}$$

Fig. 12 shows the temperature and density field of the supersonic flow past the aerofoil. The computational domain size $7C \times 4C$ with grid resolution characterized by the number of grid points along the aerofoil chord $C = 150$ with the aerofoil head positioned at $2C/3$ The. The boundary conditions are same as flow over cylinder case. Other parameters include the far-field temperature $T_\infty = 0.25$, the angle of attack of the aerofoil $\alpha = 0^o$ and shifted lattice velocity $U = (0.4u_\infty, 0)$. In the present case, the numerical solution is only converged when the shifted lattice velocity in the range of $0.4u_\infty \le U \le 0.6u_\infty$. Using shifted lattice velocity, the populations do not fall on lattices. Using shifted lattice velocity, the populations are resolved on the lattice nodes

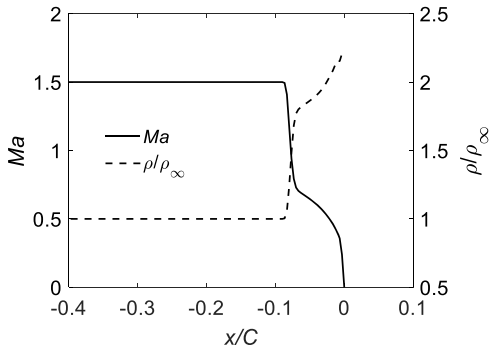


Fig. 14. Mach number and density ratio of the flow along the stagnation streamline for a supersonic flow across a NACA0012 aerofoil $Ma_\infty = 1.5,\ Re = 10^4$.

through inverse distance weighted interpolation before the streaming step. Despite our relatively low grid resolution, it is remarkable that our simulation result effectively captures the wake profile behind the aerofoil. Fig. 13 shows our LBM results compares well with other findings [36] in terms of derived pressure coefficient and skin friction coefficient along the aerofoil surface. In perspective, Latt et al. [26], Saadat et al. [21] and Frapolli et al. [30] had used much finer grid resolutions at $C = 350,\ 650,\ 800$, respectively, for comparable results.

One of the issues for high Reynolds supersonic flow simulations is numerical stability and convergence. The equilibrium populations f_i^{eq} and g_i^{eq} are to be evaluated at every grid point, numerically or analytically. Saadat et al. [21] solved the equilibrium population f_i^{eq} and g_i^{eq} analytically, but g_i^{eq} being negative past a threshold Mach number (Fig. 1) may lead to numerical divergence at higher Mach numbers, which requires a fine mesh at $C = 650$ for stability. In contrast, Latt et al. [26] solved for the equilibrium population f_i^{eq} numerically on a mesh resolution of $C = 350$ constrained by a large set of 13 equations, which leads to problems of numerical stiffness or other convergence issues. Frapolli et al. [30] also solved equilibrium population f_i^{eq} numerically using a high order lattice on a fine mesh at $C = 800$ and a set of 5 equations of constraint.

In our proposed method, only 3 equations of constraint for equilibrium population g_i^{eq} is required. Accurate solution is achieved at a low mesh resolution of only C=150 for flows over an aerofoil NACA0012 at $Ma_\infty = 1.5$ and $Re = 10^4$. Fig. 14 shows Mach number and density ratio profiles along the stagnation streamline for a supersonic flow across a NACA0012 aerofoil ($Ma_\infty = 1.5,\ Re = 10^4$). A sharp shock front is observed near the aerofoil head at a shorter distance compared to cases shown in parts B and C, and the Mach number is significantly reduced near the aerofoil head.

In order to test the stability of proposed LBM, we investigate a similar supersonic flow but at higher Reynolds numbers. As far as our knowledge, the highest Reynolds number reported in literature [21,26,30] for same problem using LBM is $Re = 10^4$. Fig. 15 shows snapshots of unsteady Mach number fields for $Re = 10^3,\ 10^4,\ 10^6,\ 10^7$ at a finer grid resolution $C = 300$. We use a denser mesh in order to capture more detail the wake as the Reynolds number is higher. The result shows that the flow wake is laminar at $Re = 10^3$, but becomes increasingly oscillatory at higher Reynolds numbers. An identical strong bow shock is observed ahead of the aerofoil independent of *Re*. Through this testing, it proves that our LBM methodology can be used for quite high Reynolds and high Mach numbers flow.

The current two-dimensional simulation does not model turbulence behaviour at high Reynolds number, since current methods are not expected to yield reliable solutions. Instead, the present study is focuses on the numerical methods that yield high fidelity solutions for supersonic flows corroborated by results shown in Figs. 8, 10 and 13.

The present LBM method can be straightforwardly extended from two to three dimensions. Here we employ analytical and numerical solutions for equilibrium populations f_i^{eq} and g_i^{eq} respectively. For a 3D simulation, f_i^{eq} is

$$f_i^{eq} = \rho \Phi_{ix} \Phi_{iy} \Phi_{iz}, \tag{21}$$

while g_i^{eq} remains unchanged as Eq. (16), where $\alpha = x, y,\ z$. Lagrange multipliers χ and ξ_α are numerically obtained by solving a set of 4 equations, 3 from Eq. (16) and 1 from Eq. (9). The D3Q39 lattice is generally appropriate for 3D simulations [26].

Fig. 16 shows snapshots of the correction coefficients σ for $Re = 10^3$,

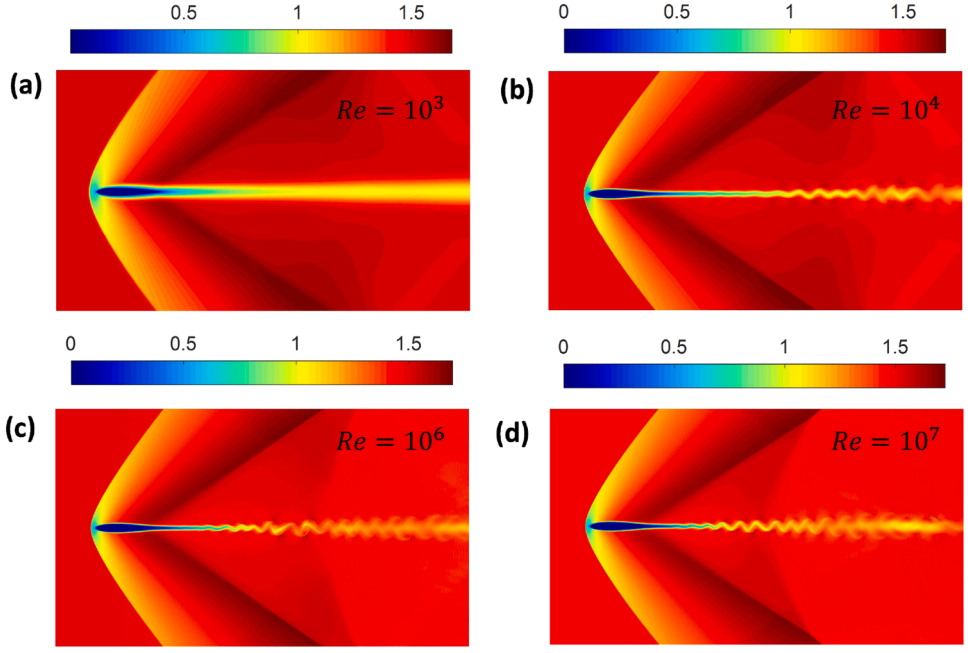


Fig. 15. Mach number fields for supersonic flow past a NACA0012 aerofoil for $Ma_\infty = 1.5$, $C = 300$, varying Reynolds numbers *Re* (a) 10^3, (b) 10^4, (c) 10^6 and (d) 10^7.

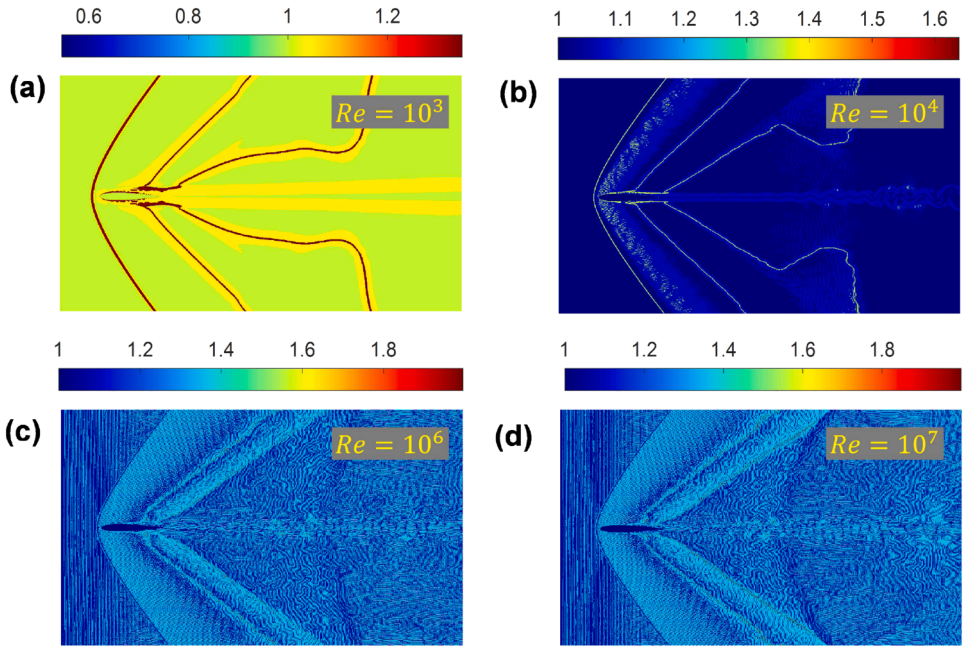


Fig. 16. Correction coefficients σ of supersonic flow past a NACA0012 aerofoil for $Ma_\infty = 1.5$, $C = 300$, varying Reynolds numbers Re (a) 10^3, (b) 10^4, (c) 10^6 and (d) 10^7.

$10^4, 10^6, 10^7$. At the shock regions, the correction coefficient deviates significantly from the noncorrecting value $\sigma = 1$ so that the instability can be dampen. As Reynolds number increases, the instability rises. Hence, the correction coefficient almost appears in the whole computational domain. As shown in Fig. 14, the Mach number and density of flow are stable without any oscillation at the shock for case $Re = 10^4$. This provide the evidence that the local correction coefficient is necessary to damp the numerical oscillation and stabilize the shock automatically, especially at high Reynolds number flow.

4. Conclusion

We presented an LBM model for compressible flow which uses high accurate estimation of local equilibria f_i^{eq} and g_i^{eq}analytically and numerically to avoid the instability due to the approximation error when the flow is at extreme condition such as high Reynolds and high Mach numbers. We adopted stabilization technique using local relaxation time and concept of shifted lattice. The model is shown to be accurate and robust through test examples including shock and expansion waves in Sod shock tube, supersonic flow over the circular cylinder, transonic and supersonic flows over the NACA0012 aerofoil. Remarkably, despite modest grid resolution, the current LBM model achieves high fidelity for simulations of supersonic flows at high Reynolds and Mach numbers ($Ma_\infty = 1.5,\ Re = 10^7$). This work can be extended straightforwardly to three-dimensions Cases 12).

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Si Bui Quang Tran: Writing – original draft, Writing – review & editing. **Fong Yew Leong:** Conceptualization, Writing – original draft, Writing – review & editing. **Quang Tuyen Le:** Conceptualization. **Duc Vinh Le:** Writing – original draft, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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